Computer Network

3rd September 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Start your answer to each question on a new page.

Answer ALL questions

Marks

Question 1

a) Compare the OSI model structure, with the TCP/IP model in computer network. You should clearly state the function of each layer and how the layers in the two models correspond. **10**

Total 10 Marks

Question 2

a) List FOUR (4) network topologies used in Ethernet and fast Ethernet and explain each of the topologies. Name **ONE (1)** topology which is becoming obsolete **and** explain the reason behind it. **10**

Total 10 Marks

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Marks

Question 3

a) Draw a network logical *diagram* to represent the following scenario. **10**

You are tasked with designing a network diagram for a small LAN (Local Area Network) in an office environment. The office consists of three departments: Sales, Marketing and Administration, each with its own set of requirements and devices.

Requirements:
1. Sales Department:
   - 10 desktop computers for sales representatives
   - 1 printer shared among sales representatives
   - Access to a shared sales database hosted on a server

2. Marketing Department:
   - 8 desktop computers for marketing executives
   - 1 printer shared among marketing executives
   - Access to design software installed on a dedicated workstation

3. Administration Department:
   - 6 desktop computers for administrative staff
   - 1 printer shared among administrative staff
   - Access to accounting software installed on a dedicated server

Additional Information:
- All departments require internet access connection.
- Wireless access is needed in common areas for guests and mobile devices.
- Security measures must be implemented to safeguard data and network resources.
- Scalability and flexibility for future expansion should be considered in the design.

Total 10 Marks

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Marks

Question 4

a) Justify the usage of the **FOUR (4)** network components used in your answer in the Question 3a. **4**

b) You have been tasked with troubleshooting a network issue reported by users in the Sales department. Users are experiencing intermittent connectivity problems and slow network performance when accessing the shared sales database hosted on a server. **6**

Outline **THREE (3)** types of information you will gather about the problem to start your troubleshooting. You should specify the source of information and/or any tools you use to gather this information.

Total 10 Marks

Question 5

a) Explain **THREE (3)** factors you should consider when deciding on the choice of network media. **6**

b) Compare and contrast CAT5e and CAT6a cabling in a LAN with respect to the following factors: **4**

1. Bandwidth
2. Speed
3. Structure
4. Application Area

Total 10 Marks

Question 6

a) Imagine a scenario where a small office network consists of a single router connecting multiple computers to the internet. One day, the router fails, causing a complete loss of internet connectivity for all users in the office.

i) Explain which principle of network design can provide the solution for this issue. Give a justification for your answer. **2**

ii) Outline **ONE (1)** specific device that could be implemented to provide -the solution in the network and describe how it would help mitigate the impact of the router failure. **2**

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Marks

b) Explain the following computer network threats and state **ONE (1)** example for each of the security features or controls which can detect/mitigate them. **6**

- DOS
- Packet sniffing
- Phishing

Total 10 marks

Question 7 **10**

Below is a Risk Assessment Matrix which can be used to identify the risk level from likelihood and impact levels of a threat.

Use the matrix to identify the overall risk score for the threats shown in the table below and devise **ONE (1)** control (or resolution strategy) for each identified risk to mitigate its likelihood and reduce its impact on the network infrastructure.

	Risk Matrix	Severity				
		Insignificant	Minor	Moderate	Major	Severe
Likelihood	**Almost Certain**	Medium	High	Very High	Very High	Very High
	Likely	Medium	High	High	Very High	Very High
	Possible	Low	Medium	High	High	Very High
	Unlikely	Low	Low	Medium	Medium	High
	Rare	Low	Low	Low	Low	Medium

Risk	***Likelihood***	***Impact***	***Risk Score***	***Control***
Unauthorised access	Possible	Minor		
Insider threat	Likely	Moderate		
Hardware failure	Possible	Moderate		
Power outage	Rare	Minor		
Human error	Likely	Major		

Total 10 Marks

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Question 8

a) Name and explain **TWO (2)** different types of firewalls and compare and contrast them in respect of their principles of operation, advantages and problems. **10**

Total 10 Marks

Question 9

a) Answer the following questions about a VPN.

i) Explain what VPN stands for. Discuss **TWO (2)** primary functions of VPN and provide **ONE (1)** example of a scenario where using a VPN is particularly beneficial. **5**

ii) Which **TWO (2)** CIA properties does VPN provide in communication? **2**

iii) Name **ONE (1)** VPN protocol. **1**

b) State **TWO (2)** basic functions of a VOIP. **2**

Total 10 Marks

Question 10

a) Answer the following questions about IPv4 addressing.

i) Explain IP addressing classes and their usage in different size networks. **8**

ii) Explain subnetting and its advantage in networking. **2**

Total 10 marks

END OF PAPER

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